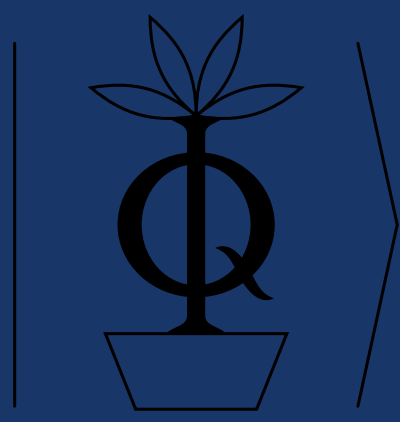


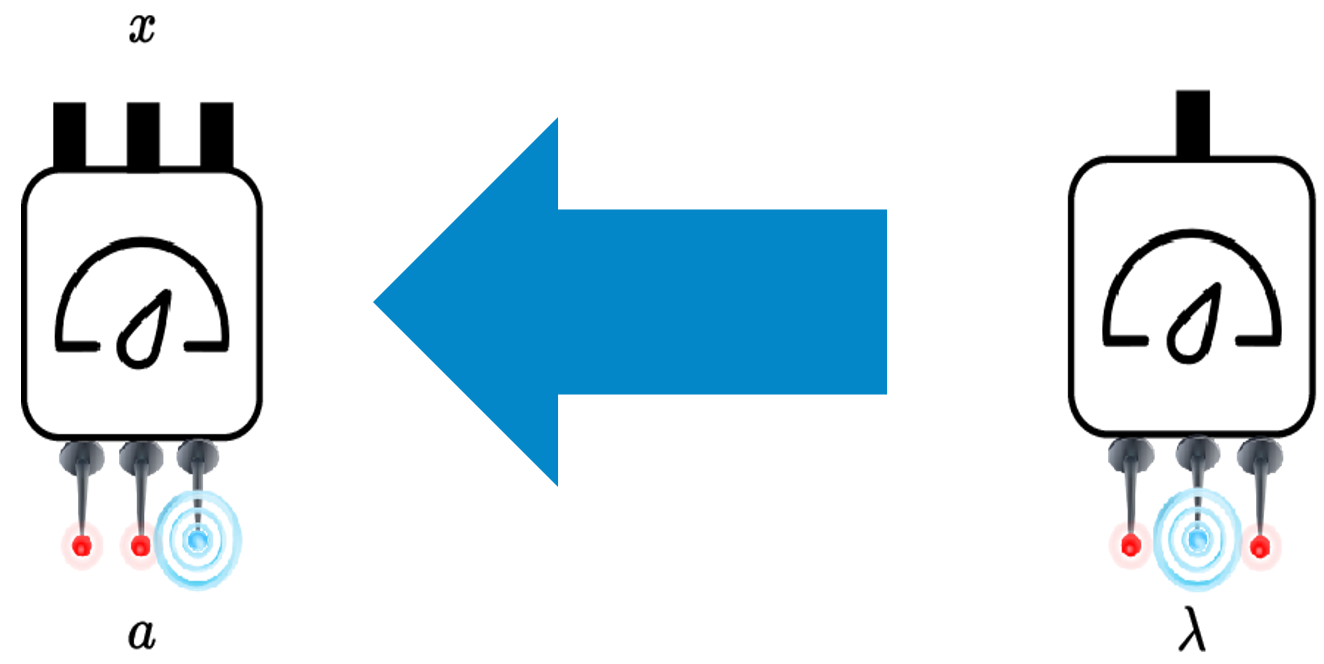
Quantum correlations cannot be reproduced with a finite number of measurements in any no-signaling theory



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Measurement Incompatibility



Definition:

$$M_{a|x} = \sum_{\lambda} p(a|x, \lambda) G_{\lambda} \iff \sum_{\vec{a} \setminus a_x} G_{\vec{a}} = M_{a|x}.$$

Objective: Simulate measurement assemblage $\mathcal{M} := \{M_{a|x}\}_{a,x}$ using one parent POVM $\{G_{\lambda}\}_{\lambda}$ and classical post-processing.

$\implies$ **Reduction of n measurements to 1 measurement.**

- Post-processing $p(a|x, \lambda)$ can be reduced to deterministic responses $D_{\lambda}(a|x)$.
- $\{G_{\vec{a}}\}_{\vec{a}}$ is a parent POVM with outcomes $[a_x=1, a_x=2, \dots, a_x=n]$.

Example:

Noise model:

$$M_{\pm|1}^{\eta} = \frac{1}{2}(\mathbb{1} \pm \eta \sigma_X), \quad M_{\pm|2}^{\eta} = \frac{1}{2}(\mathbb{1} \pm \eta \sigma_Z). \quad M_{a|x}^{\eta} = \eta M_{a|x} + (1 - \eta) \text{Tr}[M_{a|x}] \frac{\mathbb{1}}{d}.$$

- Find maximal visibility parameter $\eta \in [0, 1]$ such that $\mathcal{M}_1, \mathcal{M}_2$ are jointly measurable.
- Candidate parent POVM: $G_{i,j}^{\eta} = \frac{1}{4}(\mathbb{1} + \eta(i\sigma_X + j\sigma_Z))$ where $i, j = \pm 1$.
- Leads to maximal $\eta^* = \frac{1}{\sqrt{2}}$, which can be shown to be optimal.

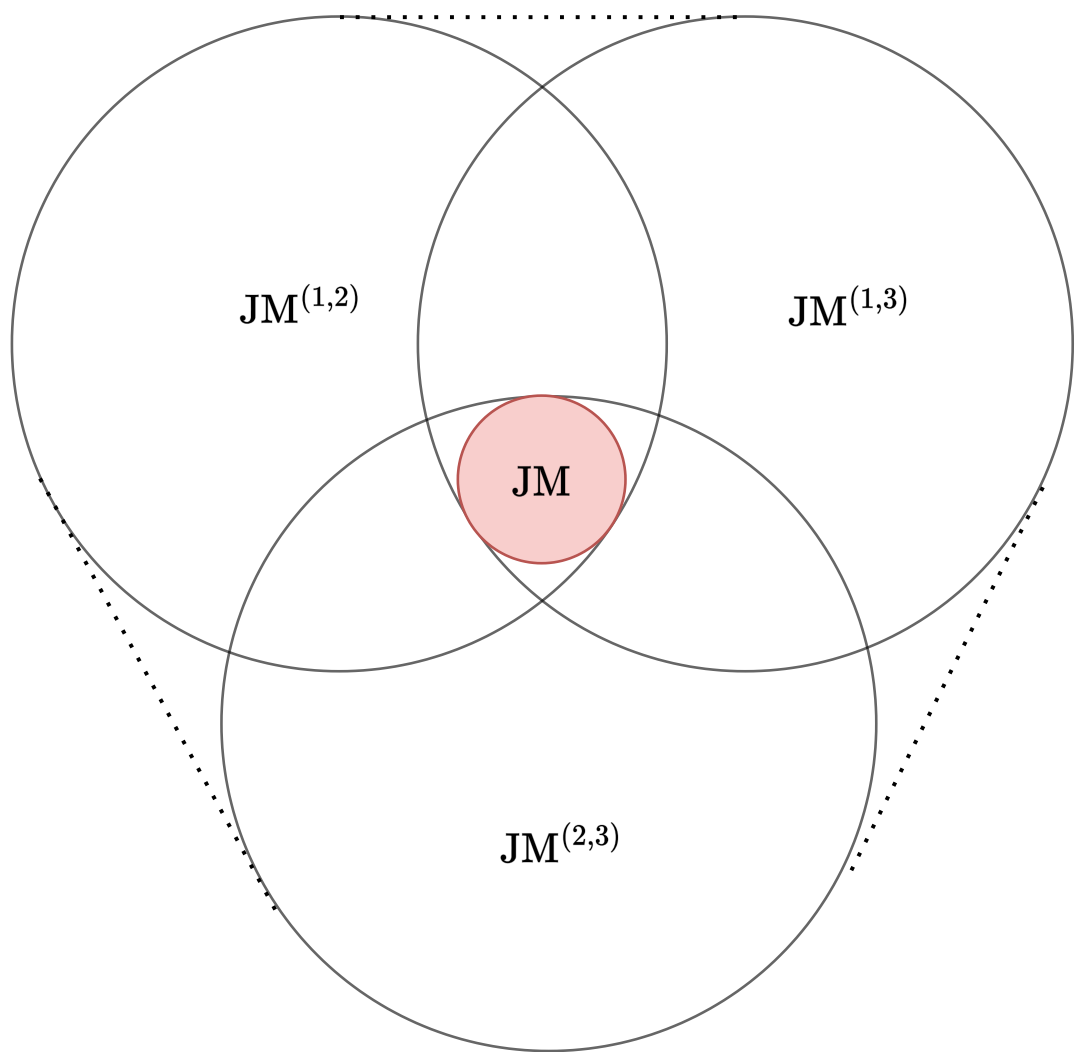
Generalized Incompatibility

Main Questions:

What does it mean for an assemblage $\mathcal{M}$ to contain *genuinely* n measurements?

Compatibility structures: [1]

- Full joint measurability JM
- One JM pair: $JM^{(1,2)}, JM^{(2,3)}, JM^{(1,3)}$.
- Pairwise JM: $JM^{(1,2)} \cap JM^{(2,3)} \cap JM^{(1,3)}$
- 2-wise compatible: $\text{Conv}(JM^{(1,2)}, JM^{(1,3)}, JM^{(2,3)})$



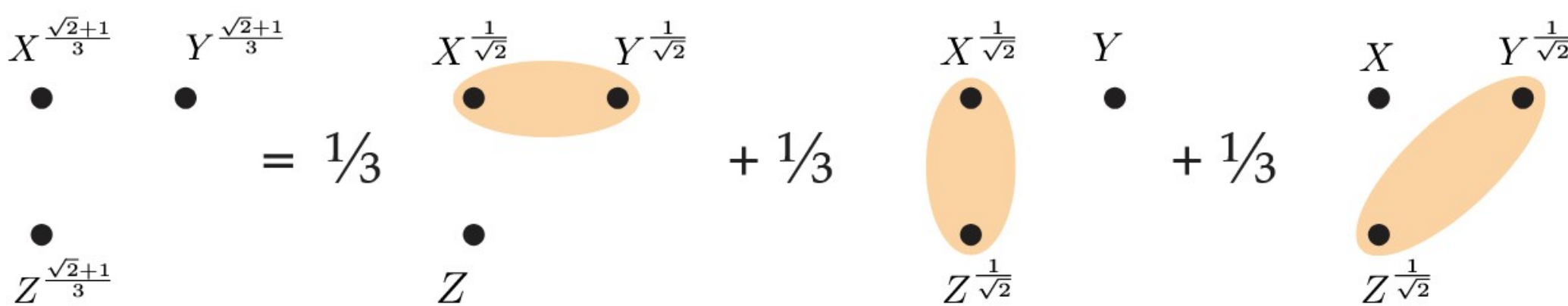
Definition of (n-1) compatible:

$$M_{a|x} = \sum_{(t_1, t_2 \neq t_1) \in \mathcal{T}} p(t_1, t_2) J_{a|x}^{(t_1, t_2)}.$$

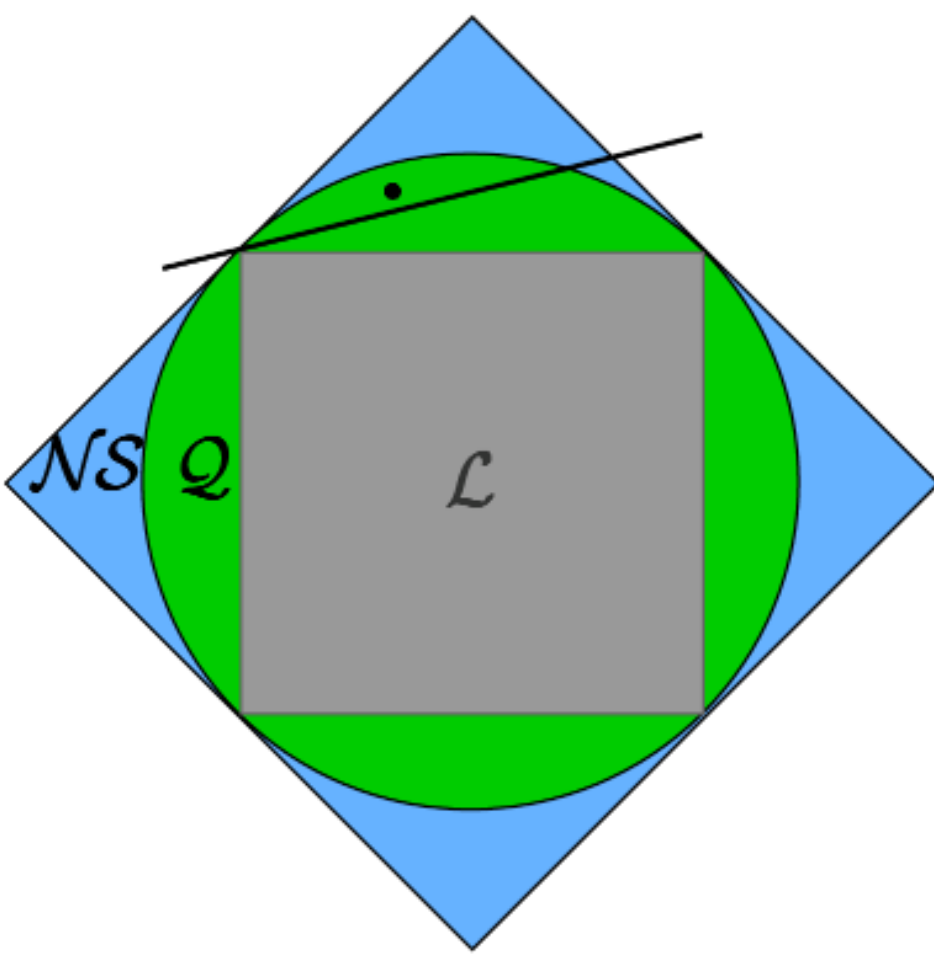
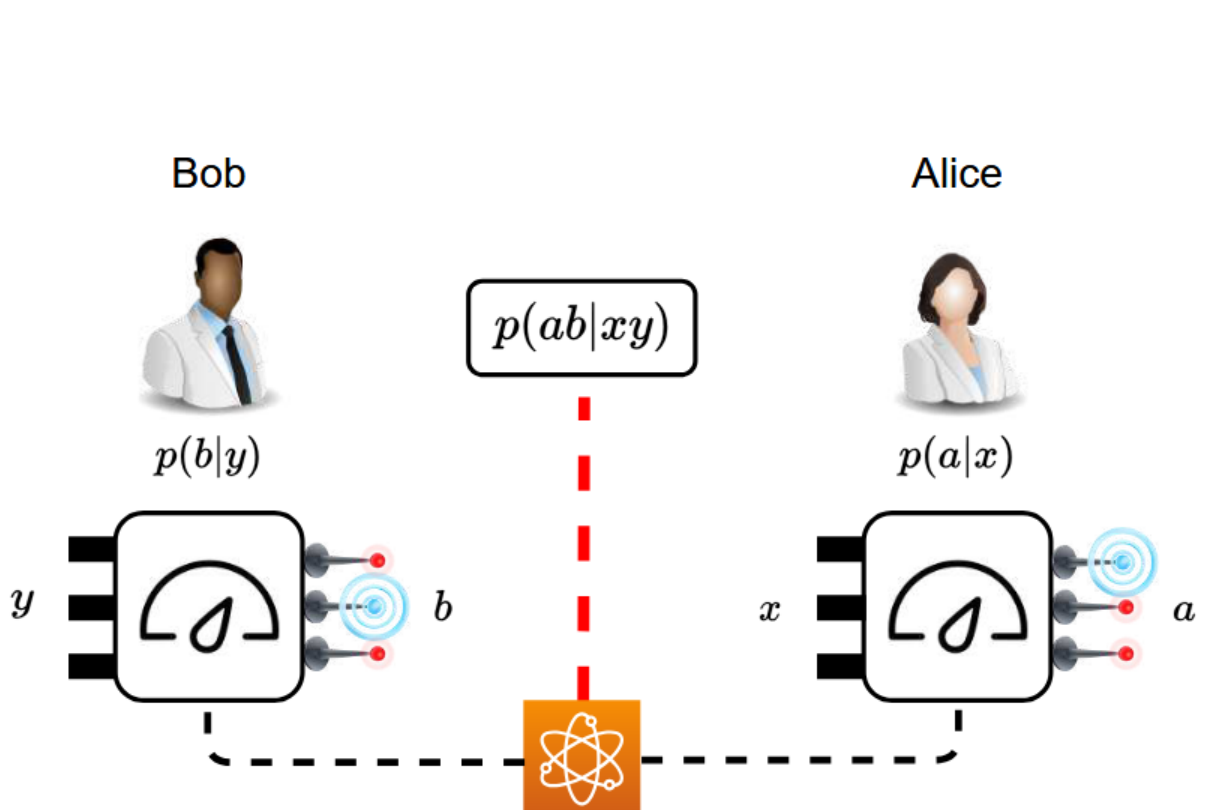
Example:

$$M_{\pm|s}^{\eta} = \frac{1}{2}(\mathbb{1} \pm \eta \sigma_s) \text{ with } s \in \{X, Y, Z\}.$$

- JM: $\eta \leq 1/\sqrt{3} \approx 0.58$.
- Pairwise JM: $\eta \leq 1/\sqrt{2} \approx 0.71$.
- 2-wise compatible: $\eta \leq \frac{\sqrt{2}+1}{3} \approx 0.80$.



Measurement Incompatibility in Bell scenarios



Bell inequality: $\sum_{a,b,x,y} C_{abxy} p(ab|xy) \stackrel{\mathcal{L}}{\leq} L \stackrel{\mathcal{Q}}{\leq} Q \stackrel{\mathcal{NS}}{\leq} \text{NS}$

JM implies LHV in Quantum Theory:

$$A_{a|x} = \sum_{\lambda} p(a|x, \lambda) G_{\lambda} \implies p(ab|xy) = \text{Tr}[(A_{a|x} \otimes B_{b|y}) \rho] = \sum_{\lambda} p_{\lambda} p(a|x, \lambda) p(b|y, \lambda) \forall B, \rho.$$

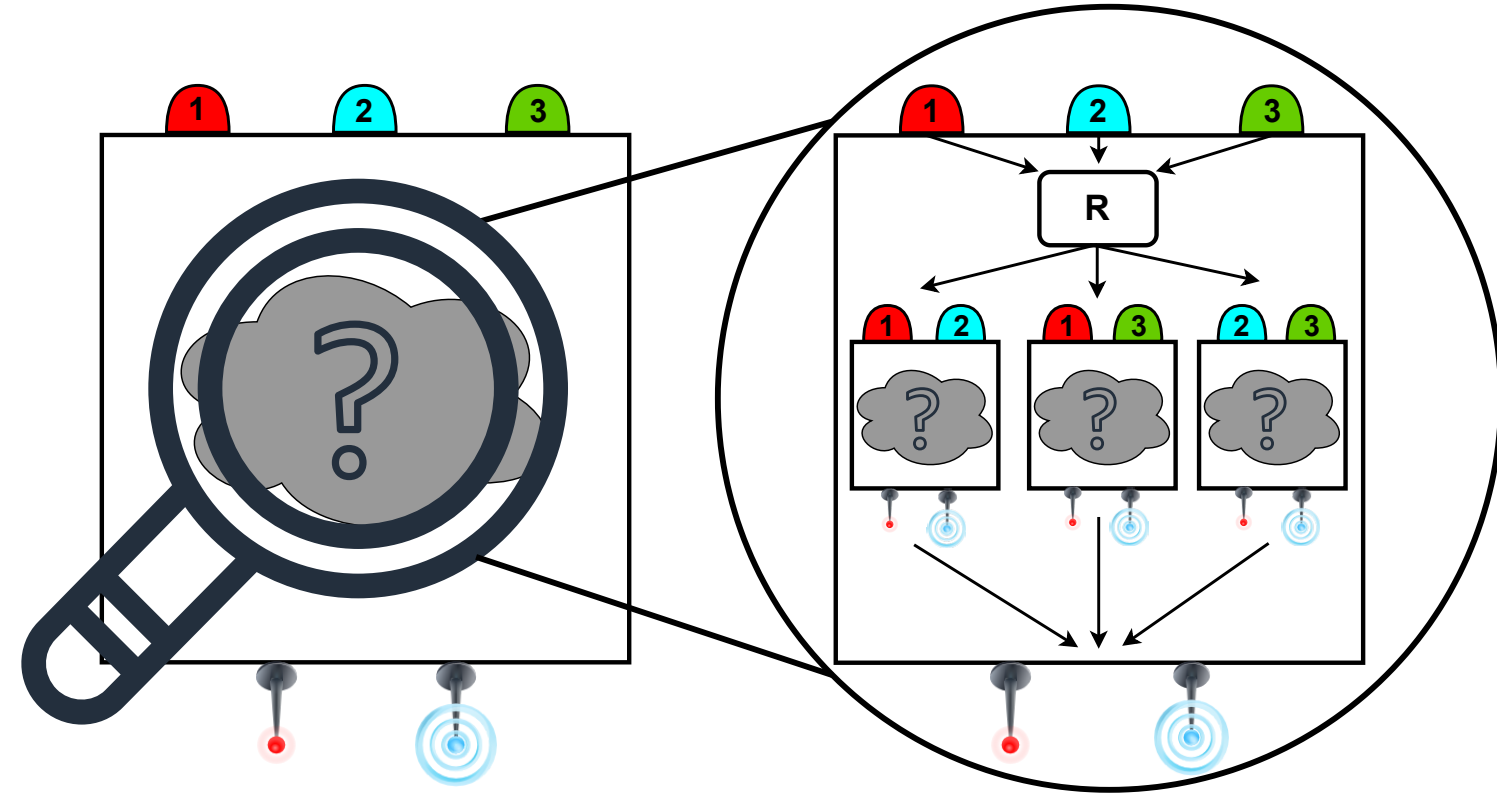
JM implies LHV in any No-signaling Theory:

$$p(ab|xy) = \sum_{\lambda} p(a|x, \lambda) p(b|y, \lambda) = \sum_{\lambda} p(\lambda) p(a|x, \lambda) p(b|y, \lambda)$$

$\implies$ **Violation of a Bell inequality is a device-independent and theory-independent certificate of incompatibility.**

Main Result

Objective: Show that any finite number n of quantum measurements can be certified device-independently and theory-independently.



To Do: Design Bell inequality that can rule out correlations of the form $p(ab|xy) = \sum_{(s,t \neq s) \in \mathcal{T}} p_{(s,t)}^{(s,t)}(ab|xy)$, where $p_{(s,t)}^{(s,t)}(ab|xy)$ describes a behaviour $P^{(s,t)}$ that is local for settings (s, t) .

Collins-Gisin notation

	P(A ₁)	P(A ₂)	P(A ₃)
P(B ₁)	P(A ₁ B ₁)	P(A ₂ B ₁)	P(A ₃ B ₁)
P(B ₂)	P(A ₁ B ₂)	P(A ₂ B ₂)	P(A ₃ B ₂)
P(B ₃)	P(A ₁ B ₃)	P(A ₂ B ₃)	P(A ₃ B ₃)

$$P(A_1) := p_A(a=0|x=1), \quad P(A_1B_1) := p(a=0, b=0|x=1, y=1)$$

CHSH inequality

$$\begin{vmatrix} & -1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix} \leq 0$$

$$p(00|11) + p(00|12) + p(00|21) - p(00|22) - p_A(0|1) - p_B(0|1) \leq 0.$$

The case of 3 measurements [2]

A violation of the inequality $M_{3322} = \begin{vmatrix} & -2 & 0 & 0 \\ -2 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ 0 & 1 & -1 & 0 \end{vmatrix} \leq 0$, **implies a simultaneous violation of:**

$$M_{3322}^{(1,2)} = \begin{vmatrix} & -1 & 0 & 0 \\ -1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{vmatrix} \leq 0, \quad M_{3322}^{(1,3)} = \begin{vmatrix} & -1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{vmatrix} \leq 0, \quad M_{3322}^{(2,3)} = \begin{vmatrix} & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{vmatrix} \leq 0.$$

Follows directly from No-signaling constraints, such as:

$$[P(A_2B_1) - P(B_1)] + [P(A_2B_2) - P(B_2)] + [P(A_1B_3) - P(A_1)] - P(A_2B_3) \leq 0, \\ \implies \text{Hence, Bob's measurements need to be genuinely triplewise incompatible.}$$

Theorem: For any $n \geq 2$, the violation of the M_{nn22} inequality certifies genuinely n measurements in any NS theory.

$\implies$ **Crucially, the M_{nn22} inequality can be violated with n quantum measurements.**

Proof Steps:

- Violation of M_{nn22} implies violation of $M_{(n-1)(n-1)22}$
- Bob's first measurement is pairwise incompatible with any other measurements
- Perform induction step
- M_{nn22} certifies pairwise incompatibility between all measurement pairs
- Due to linearity, convex combinations can be ignored

Quantum violation:

Take $|\Psi_{\epsilon}\rangle = \sqrt{\frac{1-\epsilon^2}{n-1}} \sum_{k=1}^{n-1} |kk\rangle + \epsilon |nn\rangle$, and appropriate projective measurements [3] such that

$$M_{nn22}(Q) = \epsilon^2(q_0^2n - (n-1)), \text{ with } \epsilon^2 = \frac{1-q_0^2}{1+[(n-1)^2-1]q_0^2} \text{ and } 0 \leq q_0 \leq 1.$$

We find $M_{nn22}(Q) > 0$ whenever $\sqrt{\frac{n-1}{n}} < q_0 < 1$. **Examples:** $M_{3322} \approx 0.0239$, $M_{5522} \approx 0.0035$.

Conclusions and Open Questions

- Any finite number n of quantum measurements can be certified device-independently.
- Any No-signaling theory restricted to $(n-1)$ *genuine measurements* fails to reproduce quantum theory.
- It takes an unbounded number of measurements to reproduce the predictions of quantum theory.

Open questions:

- Can the experimental demands be lowered? $\implies$ Larger violations for other inequalities possible?
- What is the minimal dimension $d(n)$ needed to certify theory-independently n genuine measurements?
- What are the implications for the state-space? $\implies$ Is an unbounded number of pure states necessary?
- Other degrees of freedom that can be tested? **Outcomes** [4], **entanglement structure** [5], **multipartiteness of sources** [6], **others?**

References

- [1] M.T. Quintino *et al.*, PRL **123**, 180401 (2019).
- [2] N. Brunner *et al.*, New. J. Phys **7**, 88 (2005).
- [3] T. Vértesi *et al.*, PRL **104**, 060401 (2010).
- [4] M. Kleinmann, A. Cabello, PRL **117**, 150401 (2016).
- [5] D. Collins *et al.*, PRL **88**, 170405 (2002).
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